



EAST WEST UNIVERSITY

Assignment 02

Course Details

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Submitted To

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1. Consider the wave function

$$\Psi(x,t) = Ae^{-2|x|} e^{-i\omega t}$$

Where A, λ and ω are positive real constants.

(a) Normalize Ψ

(b) Determine the expectation values of x and x^2 .

Soln:

a.

Normalize the wave function by requiring the integral of $|\Psi(x,t)|^2$ over all x to be 1.

$$1 = \int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx$$

$$= \int_{-\infty}^{\infty} \Psi(x,t) \Psi^*(x,t) dx$$

$$= \int_{-\infty}^{\infty} (Ae^{-2|x|} e^{-i\omega t}) (Ae^{-2|x|} e^{i\omega t}) dx$$

$$= A^2 \int_{-\infty}^{\infty} e^{-2\lambda|x|} dx$$

$$= A^2 \left[\int_{-\infty}^0 e^{2\lambda x} dx + \int_0^{\infty} e^{-2\lambda x} dx \right]$$

$$= A^2 \left(\int_{-\infty}^0 e^{2\lambda x} dx + \int_0^{\infty} e^{-2\lambda x} dx \right)$$

$$= A^2 \left(\frac{1}{2\lambda} e^{2\lambda x} \Big|_{-\infty}^0 + \frac{1}{-2\lambda} e^{-2\lambda x} \Big|_0^{\infty} \right)$$

$$= A^2 \left[\frac{1}{2\lambda} (1-0) + \frac{1}{-2\lambda} (0-1) \right]$$

$$= A^2 \left(\frac{1}{\lambda} \right)$$

$$\therefore A = \sqrt{\lambda}$$

With this value of A , the wave function becomes,

$$\Psi(x,t) = \sqrt{\lambda} e^{-\lambda|x|} e^{-i\omega t}$$

According to Born's interpretation, the probability distribution for the particle's position at a time t is,

$$|\Psi(x,t)|^2 = \Psi(x,t) \Psi^*(x,t)$$

$$= (\sqrt{\lambda} e^{-\lambda|x|} e^{-i\omega t}) (\sqrt{\lambda} e^{-\lambda|x|} e^{i\omega t})$$

$$= \lambda e^{-2\lambda|x|}$$

Soln:

b Now using the value from 'a' we get,

$$\begin{aligned}\langle x \rangle &= \frac{\int_{-\infty}^{\infty} x |\Psi(x,t)|^2 dx}{\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx} \\&= \frac{\int_{-\infty}^{\infty} x |\Psi(x,t)|^2 dx}{1} \\&= \int_{-\infty}^{\infty} x |\Psi(x,t)|^2 dx \\&= \int_{-\infty}^{\infty} x \Psi(x,t) \Psi^*(x,t) dx \\&= \int_{-\infty}^{\infty} x (\sqrt{2} e^{-2|x|} e^{-i\omega t}) (\sqrt{2} e^{-2|x|} e^{i\omega t}) dx \\&= 2 \int_{-\infty}^{\infty} x e^{-2|x|} dx \\&= 0\end{aligned}$$

This last integration is zero because the integral is of an odd function over a symmetric interval.

$$\begin{aligned}\langle x^2 \rangle &= \frac{\int_{-\infty}^{\infty} x^2 |\Psi(x,t)|^2 dx}{\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx} \\&= \frac{\int_{-\infty}^{\infty} x^2 |\Psi(x,t)|^2 dx}{1}\end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} x^2 |\Psi(x,t)|^2 dx \\
&= \int_{-\infty}^{\infty} x^2 \Psi(x,t) \Psi^*(x,t) dx \\
&= \int_{-\infty}^{\infty} x^2 (\sqrt{\lambda} e^{-\lambda|x|} e^{-i\omega t}) (\sqrt{\lambda} e^{-\lambda|x|} e^{i\omega t}) dx \\
&= \lambda \int_{-\infty}^{\infty} x^2 e^{-2\lambda|x|} dx \\
&= 2\lambda \int_0^{\infty} x^2 e^{-2\lambda x} dx \\
&= 2\lambda \int_0^{\infty} x^2 e^{-2\lambda x} dx \\
&= 2\lambda \int_0^{\infty} \frac{1}{4} \frac{\partial^2}{\partial \lambda^2} (e^{-2\lambda x}) dx \\
&= \frac{\lambda}{2} \frac{d^2}{d\lambda^2} \int_0^{\infty} e^{-2\lambda x} dx \\
&= \frac{\lambda}{2} \frac{d^2}{d\lambda^2} \left(-\frac{1}{2\lambda} e^{-2\lambda x} \Big|_0^{\infty} \right) \\
&= \frac{\lambda}{2} \frac{d^2}{d\lambda^2} \left[-\frac{1}{2\lambda} (0 - 1) \right] \\
&= \frac{\lambda}{4} \frac{d^2}{d\lambda^2} \left(\frac{1}{\lambda} \right) \\
&= \frac{\lambda}{4} \left(-\frac{2}{\lambda^3} \right) \\
&= -\frac{1}{2\lambda^2} \\
\therefore \langle x^2 \rangle &= \frac{1}{2\lambda^2}
\end{aligned}$$

2. Find the first excited state of the harmonic oscillator.

Soln:

$$\Psi_1(x) = A_1 a_+ \Psi_0(x)$$

$$\Psi_1(x) = A_1 \frac{1}{\sqrt{2m\hbar\omega}} (-ip + m\omega x) \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\text{Since } -ip = -i\hbar \frac{d}{dx}$$

$$= \frac{A_1}{\sqrt{2m\hbar\omega}} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \left(-\hbar \frac{d}{dx} + m\omega x\right) e^{-\frac{m\omega}{2\hbar}x^2}$$

Taking the derivative,

$$\begin{aligned} -\hbar \frac{d}{dx} e^{-\frac{m\omega}{2\hbar}x^2} &= +\hbar \frac{m\omega}{\hbar} x e^{-\frac{m\omega}{2\hbar}x^2} \\ &= m\omega x e^{-\frac{m\omega}{2\hbar}x^2} \end{aligned}$$

So,

$$\frac{A_1}{\sqrt{2m\hbar\omega}} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \left[m\omega x e^{-\frac{m\omega}{2\hbar}x^2} + m\omega x e^{-\frac{m\omega}{2\hbar}x^2} \right]$$

$$= \frac{A_1}{\sqrt{2m\hbar\omega}} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} 2m\omega x e^{-\frac{m\omega}{2\hbar}x^2}$$

Simplifying $\frac{2m\omega}{\sqrt{2m\hbar\omega}}$,

$$\frac{2m\omega x}{\sqrt{2m\hbar\omega}} = \sqrt{\frac{2m\omega}{\hbar}} x$$

$$\text{or, } \Psi_1(x) = A_1 \sqrt{\frac{2m\omega}{\hbar}} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} x e^{-\frac{m\omega}{2\hbar}x^2}$$

Finding A_1 (Normalization)

$$\int_{-\infty}^{\infty} |\psi_1|^2 dx = 1$$

$$|A_1|^2 \left(\frac{2m\omega}{\hbar} \right) \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{2}} \int_{-\infty}^{\infty} x^2 e^{-\frac{m\omega}{\hbar} x^2} dx = 1$$

Using the standard integral:

$$\int_{-\infty}^{\infty} x^{2n} e^{-x^2/a^2} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{a}{2} \right)^{2n+1}$$

For $n=1$, with $a = \sqrt{\frac{\hbar}{m\omega}}$

$$\int_{-\infty}^{\infty} x^2 e^{-\frac{m\omega}{\hbar} x^2} dx = \sqrt{\pi} \left(\frac{a}{2} \right)^3 \cdot 2 = \frac{\sqrt{\pi}}{2} \left(\frac{\hbar}{m\omega} \right)^{\frac{3}{2}}$$

Substituting back:

$$|A_1|^2 \left(\frac{2m\omega}{\hbar} \right) \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{2}} \cdot \frac{\sqrt{\pi}}{2} \left(\frac{\hbar}{m\omega} \right)^{\frac{3}{2}} = 1$$

All terms cancel out:

$$|A_1|^2 = 1$$

$$\text{or, } A_1 = 1$$

So the wavefunction is,

$$\psi_1(x) = \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} \sqrt{\frac{2m\omega}{\hbar}} x e^{-\frac{m\omega}{2\hbar} x^2}$$

Finding E_1 ,

General energy formula for the harmonic oscillator:

$$E_n = \left(n + \frac{1}{2} \right) \hbar \omega$$

For the first excited state, $n = 1$

$$E_1 = \left(1 + \frac{1}{2}\right) \hbar \omega$$

$$\therefore E_1 = \frac{3}{2} \hbar \omega$$

3. Find the $\langle x \rangle$, $\langle p \rangle$ and $\langle x^2 \rangle$ for the ground state $\psi_0(x)$ of harmonic oscillator.

Soln:

For the ground state of the harmonic oscillator, the wave function is,

$$\psi_0(x) = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}}$$

Therefore,

$$\psi_0^*(x) = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}}$$

and hence,

$$|\psi_0(x)|^2 = \psi_0^*(x) \psi_0(x) = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{2}} e^{-\frac{m\omega x^2}{\hbar}}$$

The expectation value of x is,

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{\infty} \psi_0^*(x) x \psi_0(x) dx \\ &= \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{2}} \int_{-\infty}^{\infty} x e^{-\frac{m\omega x^2}{\hbar}} dx \end{aligned}$$

Here,

$$x e^{-\frac{m\omega x^2}{\hbar}}$$

is an odd function. Therefore,

$$\int_{-\infty}^{\infty} x e^{-\frac{m\omega x^2}{\hbar}} dx = 0$$

Hence,

$$\langle x \rangle = 0$$

The momentum operator is,

$$\hat{p} = -i\hbar \frac{d}{dx}$$

Therefore,

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) \left(-i\hbar \frac{d}{dx} \right) \psi_0(x) dx$$

Now,

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar} x \psi_0(x)$$

$$\text{Thus, } -i\hbar \frac{d\psi_0}{dx} = im\omega x \psi_0(x)$$

Therefore,

$$\langle p \rangle = im\omega \int_{-\infty}^{\infty} x |\psi_0(x)|^2 dx$$

Since $x |\psi_0(x)|^2$ is an odd function,

$$\int_{-\infty}^{\infty} x |\psi_0(x)|^2 dx = 0$$

Hence,

$$\langle p \rangle = 0$$

The expectation value of x^2 is,

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) x^2 \psi_0(x) dx$$

$$= \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{2}} \int_{-\infty}^{\infty} x^2 e^{-\frac{m\omega x^2}{\hbar}} dx$$

Using, $\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2a^{3/2}}$ where $a = \frac{m\omega}{\hbar}$

we get,

$$\langle x^2 \rangle = \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{2}} \frac{\sqrt{\pi}}{2 \left(\frac{m\omega}{\hbar} \right)^{\frac{3}{2}}}$$

Therefore,

$$\langle x^2 \rangle = \frac{\hbar}{2m\omega}$$

4. A particle of mass m is bound in the delta function well $V(x) = -\alpha \delta(x)$. What is the probability that a measurement of its momentum would yield a value greater than $P_0 = m\alpha/\hbar$?

Soln: Wave function,

$$\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}, \quad E = -\frac{m\alpha^2}{2\hbar^2}$$

$$\psi(x, t) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2} e^{-iEt/\hbar}$$

The momentum space wave function,

$$\phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \frac{\sqrt{m\alpha}}{\hbar} e^{-iEt/\hbar} \int_{-\infty}^{\infty} e^{-ipx/\hbar} e^{-m\alpha|x|/\hbar^2} dx$$

$$= \sqrt{\frac{2}{\pi}} \frac{p_0^{3/2} e^{-iEt/\hbar}}{p^2 + p_0^2}$$

The probability becomes,

$$\begin{aligned} & \frac{2}{\pi} p_0^3 \int_{p_0}^{\infty} \frac{1}{(p^2 + p_0^2)} dp \\ &= \frac{1}{\pi} \left[\frac{pp_0}{p^2 + p_0^2} + \tan^{-1} \left(-\frac{p}{p_0} \right) \right] \Big|_{p_0}^{\infty} \\ &= \frac{1}{4} - \frac{1}{2\pi} \\ &= 0.0908 \end{aligned}$$

5. Find the eigenfunctions and eigenvalues of momentum operator.

Soln:

The momentum operator in one dimensional position space is, $\hat{p} = -i\hbar \frac{d}{dx}$

For an eigenfunction $\Psi(x)$, the eigenvalue eqn is,

$$\hat{p}\Psi(x) = p\Psi(x)$$

Substituting the momentum operator,

$$-i\hbar \frac{d\Psi(x)}{dx} = p\Psi(x)$$

$$\text{Now, } \frac{d\Psi(x)}{dx} = \frac{i p}{\hbar} \Psi(x)$$

Therefore,

$$\frac{1}{\Psi(x)} d\Psi(x) = \frac{i p}{\hbar} dx$$

Integrating both sides,

$$\int \frac{1}{\Psi(x)} d\Psi(x) = \frac{i p}{\hbar} \int dx$$

Hence,

$$\ln \Psi(x) = \frac{i p x}{\hbar} + C$$

Taking the exponential of both sides,

$$\Psi(x) = A e^{i p x / \hbar}$$

Where A is a constant.

For delta function normalization,

$$\int_{-\infty}^{\infty} \Psi_{p'}^*(x) \Psi_p(x) dx = \delta(p - p')$$

$$A = \frac{1}{\sqrt{2\pi\hbar}}$$

Therefore, the normalized eigenfunction is,

$$\Psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i p x / \hbar}$$

Thus the eigenfunctions are,

$$\Psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i p x / \hbar}$$

and the corresponding eigenvalues are,

$$p \in (-\infty, \infty)$$

So, the momentum operator has a continuous spectrum.

6. Prove the following commutator :

(a) $[AB, C] = A[B, C] + [A, C]B$

b. $[x^n, p] = i\hbar n x^{n-1}$

Soln:

a. By the definition of the commutator,

$$[AB, C] = ABC - CAB$$

Now, we need to show that this is equal to,

$$A[B, C] + [A, C]B$$

Consider the right hand side,

$$A[B, C] + [A, C]B$$

Using the definition,

$$[B, C] = BC - CB$$

and $[A, C] = AC - CA$

we get,

$$\begin{aligned} A[B, C] + [A, C]B &= A(BC - CB) + (AC - CA)B \\ &= ABC - ACB + ACB - CAB \end{aligned}$$

Expanding the brackets $= ABC - CAB$

But from the definition of the commutator,

$$ABC - CAB = [AB, C]$$

Hence, $[AB, C] = A[B, C] + [A, C]B$
[Proved]

Soln:
b.

Let the commutator operate on an arbitrary wave function $\psi(x)$.

By definition,

$$[x^n, p]\psi = x^n p\psi - p x^n \psi$$

We know, $p = -i\hbar \frac{d}{dx}$

Now,

$$[x^n, p]\psi = x^n \left(-i\hbar \frac{d\psi}{dx}\right) - \left(-i\hbar \frac{d\psi}{dx}\right) (x^n \psi)$$

Therefore,

$$[x^n, p]\psi = -i\hbar x^n \frac{d\psi}{dx} + i\hbar \frac{d}{dx} (x^n \psi)$$

Now applying the product rule,

$$\frac{d}{dx} (x^n \psi) = x^n \frac{d\psi}{dx} + \psi \frac{d}{dx} (x^n)$$

$$\text{or, } \frac{d}{dx} (x^n \psi) = x^n \frac{d\psi}{dx} + n x^{n-1} \psi$$

Now,

$$\begin{aligned} [x^n, p]\psi &= -i\hbar x^n \frac{d\psi}{dx} + i\hbar \left(x^n \frac{d\psi}{dx} + n x^{n-1} \psi \right) \\ &= -i\hbar x^n \frac{d\psi}{dx} + i\hbar x^n \frac{d\psi}{dx} + i\hbar n x^{n-1} \psi \end{aligned}$$

$$\text{or, } [x^n, p]\psi = i\hbar n x^{n-1} \psi$$

$$\text{or, } [x^n, p]\psi = i\hbar n x^{n-1}$$

[Proved]

7. Starting from the Rodrigues formula, derive the orthonormality condition for Legendre polynomials:

$$\int_{-1}^1 P_l(x) P_{l'}(x) dx = \left(\frac{2}{2l+1} \right) \delta_{ll'}$$

Soln: The Rodrigues formula for Legendre polynomials is,

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2-1)^l$$

considering, $I = \int_{-1}^1 P_l(x) P_{l'}(x) dx$

Substituting the Rodrigues formula,

$$I = \frac{1}{2^l l!} \int_{-1}^1 P_{l'}(x) \frac{d^l}{dx^l} (x^2-1)^l dx$$

Integrating by parts l times, the boundary terms vanish because,

$$(x^2-1)^l = 0 \quad \text{at } x = \pm 1$$

Thus,
$$I = \frac{(-1)^l}{2^l l!} \int_{-1}^1 (x^2-1)^l \frac{d^l P_{l'}(x)}{dx^l} dx$$

Assume $l > l'$. Since $P_{l'}(x)$ is a polynomial of degree l' .

$$\frac{d^l P_{l'}(x)}{dx^l} = 0$$

Therefore, $I = 0$

Hence, $\int_{-1}^1 P_l(x) P_{l'}(x) dx = 0, \quad l \neq l'$

For $l = l'$:

The highest power term of $P_l(x)$ is,

$$P_l(x) = \frac{(2l)!}{2^l (l!)^2} x^l + \dots$$

Therefore,

$$\frac{d^l P_l(x)}{dx^l} = \frac{(2l)!}{2^l l!}$$

Hence,

$$I = \frac{(-1)^l (2l)!}{2^{2l} (l!)^2} \int_{-1}^1 (x^2 - 1)^l dx$$

Since,

$$(x^2 - 1)^l = (-1)^l (1 - x^2)^l$$

we get,

$$I = \frac{(2l)!}{2^{2l} (l!)^2} \int_{-1}^1 (1 - x^2)^l dx$$

Using, $\int_{-1}^1 (1 - x^2)^l dx = \frac{2^{2l+1} (l!)^2}{(2l+1)!}$

we get, $I = \frac{(2l)!}{2^{2l} (l!)^2} \frac{2^{2l+1} (l!)^2}{(2l+1)!}$

$$I = \frac{2(2l)!}{(2l+1)!}$$

Therefore, $I = \frac{2}{2l+1}$

Combining both cases,

$$\int_{-1}^1 P_l(x) P_{l'}(x) dx = \frac{2}{2l+1} \delta_{ll'}$$

[Proved]

8. Prove the following commutator relation:

$$[L_z, x] = i\hbar y, \quad [L_z, y] = -i\hbar x, \quad [L_z, z] = 0$$

$$[L_z, p_x] = i\hbar p_y, \quad [L_z, p_y] = -i\hbar p_x,$$

$$[L_z, p_z] = 0$$

Soln:

$$[L_z, x] = i\hbar y$$

Now,

$$L_z = x p_y - y p_x$$

Now,

$$[L_z, x] = [x p_y, x] - [y p_x, x]$$

Using, $[AB, C] = A[B, C] + [A, C]B$

for the first term,

$$[x p_y, x] = x[p_y, x] + [x, x]p_y$$

Now,

$$[p_y, x] = 0 \quad \text{and} \quad [x, x] = 0$$

Therefore, $[x p_y, x] = 0$

For the second term,

$$[y p_x, x] = y[p_x, x] + [y, x]p_x$$

Since,

$$[p_x, x] = -[x, p_x] = -i\hbar$$

and $[y, x] = 0$

We get $[y p_x, x] = -i\hbar y$

Hence,

$$[L_z, x] = 0 - (-i\hbar y)$$

Therefore,

$$[L_z, x] = i\hbar y$$

[Proved]

$$[L_z, \psi] = -i\hbar \chi$$

We have,

$$[L_z, \psi] = [\chi p_y - y p_x, \psi]$$

Therefore, $[L_z, \psi] = [\chi p_y, \psi] - [y p_x, \psi]$

For the first term,

$$[\chi p_y, \psi] = \chi [p_y, \psi] + [\chi, \psi] p_y$$

Since,

$$[p_y, \psi] = -[y, p_y] = -i\hbar$$

and

$$[\chi, \psi] = 0$$

We obtain,

$$[\chi p_y, \psi] = -i\hbar \chi$$

For the second term, $[y p_x, \psi] = y [p_x, \psi] + [y, \psi] p_x$

Since,

$$[p_x, \psi] = 0, \quad [y, \psi] = 0$$

We get

$$[y p_x, \psi] = 0$$

Therefore, $[L_z, \psi] = -i\hbar \chi - 0$

Hence,

$$[L_z, \psi] = -i\hbar \chi$$

[Proved]

$$[L_z, z] = 0$$

Now,

$$[L_z, z] = [x p_y - y p_x, z]$$

Therefore,

$$[L_z, z] = [x p_y, z] - [y p_x, z]$$

For the first term,

$$[x p_y, z] = x [p_y, z] + [x, z] p_y$$

$$\text{But } [p_y, z] = 0, \quad [x, z] = 0$$

Therefore,

$$[x p_y, z] = 0$$

$$\text{Similarly, } [y p_x, z] = y [p_x, z] + [y, z] p_x$$

$$\text{Since, } [p_x, z] = 0, \quad [y, z] = 0$$

$$\text{we get, } [y p_x, z] = 0$$

$$\text{Thus, } [L_z, z] = 0 - 0$$

$$\text{Therefore, } [L_z, z] = 0$$

[Proved]

$$[L_z, P_x] = i\hbar P_y$$

Now,

$$[L_z, P_x] = [x P_y - y P_x, P_x]$$

Therefore,

$$[L_z, P_x] = [x P_y, P_x] - [y P_x, P_x]$$

For the first term,

$$[x P_y, P_x] = x [P_y, P_x] + [x, P_x] P_y$$

Since, $[P_y, P_x] = 0$

and $[x, P_x] = i\hbar$

we get, $[x P_y, P_x] = 0 + i\hbar P_y$

Thus, $[x P_y, P_x] = i\hbar P_y$

Now for the second term,

$$[y P_x, P_x] = y [P_x, P_x] + [y, P_x] P_x$$

Since, $[P_x, P_x] = 0$, $[y, P_x] = 0$

we get, $[y P_x, P_x] = 0$

Therefore, $[L_z, P_x] = i\hbar P_y - 0$

$$[L_z, P_x] = i\hbar P_y \quad [\text{Proved}]$$

$$[L_z, p_y] = -i\hbar p_x$$

Now, $[L_z, p_y] = [x p_y - y p_x, p_y]$

Therefore, $[L_z, p_y] = [x p_y, p_y] - [y p_x, p_y]$

For the first term,

$$[x p_y, p_y] = x [p_y, p_y] + [x, p_y] p_y$$

Since, $[p_y, p_y] = 0$, $[x, p_y] = 0$.

We get, $[x p_y, p_y] = 0$

For the second term, $[y p_x, p_y] = y [p_x, p_y] + [y, p_y] p_x$

Since, $[p_x, p_y] = 0$

and $[y, p_y] = i\hbar$

We get, $[y p_x, p_y] = i\hbar p_x$

Therefore, $[L_z, p_y] = 0 - i\hbar p_x$

Hence, $[L_z, p_y] = -i\hbar p_x$

[Proved]

$$[L_z, p_z] = 0$$

Now, $[L_z, p_z] = [x p_y - y p_x, p_z]$

Therefore, $[L_z, p_z] = [x p_y, p_z] - [y p_x, p_z]$

For the first term,

$$[x p_y, p_z] = x [p_y, p_z] + [x, p_z] p_y$$

Since,

$$[p_y, p_z] = 0$$

and $[x, p_z] = 0$

we get, $[x p_y, p_z] = 0$

Similarly,

$$[y p_x, p_z] = y [p_x, p_z] + [y, p_z] p_x$$

Since $[p_x, p_z] = 0$

and $[y, p_z] = 0$

we obtain, $[y p_x, p_z] = 0$

Therefore, $[L_z, p_z] = 0 - 0$

Hence, $[L_z, p_z] = 0$

[Proved]